

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 2

HSC ASSESSMENT TASK JUNE 2009

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions

NAME : _____

TEACHER : _____

QUESTION 1	QUESTION 2	QUESTION 3	TOTAL

Question 1 (17 marks)

Marks

a) Find $\int \frac{15}{x^2 + 3x - 4} dx$ 3

b) Evaluate $\int_0^{\frac{\pi}{4}} x \sin 2x dx$ 3

c) Evaluate $\int_0^{\ln 2} \frac{e^{2x}}{e^x + 1} dx$ 4

d) Two of the roots of the equation $x^3 + ax^2 + 15x - 7 = 0$ are equal and rational. Find the value of a . 3

e) The equation $x^3 - 4x + 5 = 0$ has roots α , β and δ .

Find the value of i) $\alpha^3 + \beta^3 + \delta^3$ 2

ii) $(\alpha + \beta)^2 (\alpha + \delta)^2 (\beta + \delta)^2$ 2

Question 2 (17 marks)

Marks

a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta}$ using the substitution $t = \tan \frac{\theta}{2}$ 4

b) Find $\int \frac{2x}{x^2 + 4x + 5} dx$ 4

c) If $I_n = \int_0^2 (x^3 - 8)^n dx$, where n is a positive integer, 4

show that $I_n = \frac{-24n}{3n+1} I_{n-1}$.

d) Let α be the complex root of $z^7 = 1$ with smallest positive argument.

i) Show that $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$ 2

ii) If $x^3 + ax^2 + bx + c = 0$ is a cubic equation with roots 3

$\alpha + \alpha^6$, $\alpha^2 + \alpha^5$ and $\alpha^3 + \alpha^4$,

find the values of a , b and c .

Question 3 (17 marks)

Marks

a) Evaluate $\int_2^4 \frac{dx}{x\sqrt{x-1}}$ 3

b) Find $\int \sin^4 x \cos^3 x \, dx$ 3

c) $P(x)$ is a cubic polynomial with real coefficients. 4

One zero of $P(x)$ is $1+2i$, the constant term is -15 , and $P(2) = 5$.

Find the equation of the polynomial $P(x)$.

d) The polynomial $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ 3

has a root of multiplicity 3.

Solve $P(x) = 0$.

e) Let α, β, δ be the roots of the cubic equation $x^3 + px^2 + q = 0$, 4

where p, q are real.

The equation $x^3 + ax^2 + bx + c = 0$ has roots $\frac{1}{\alpha+1}, \frac{1}{\beta+1}, \frac{1}{\delta+1}$.

Find expressions for a, b and c in terms of p and q .

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS

$$1. a. \int \frac{15}{(x+4)(x-1)}$$

$$\frac{a}{x+4} + \frac{b}{x-1} = \frac{15}{(x+4)(x-1)}$$

$$\therefore a(x-1) + b(x+4) = 15$$

$$\text{let } x=1 \quad 5b=15$$

$$b=3$$

$$x=-4 \quad -5a=15$$

$$a=-3$$

$$= \int \frac{3}{x-1} - \frac{3}{x+4} dx$$

$$= 3 \ln(x-1) - 3 \ln(x+4)$$

$$= 3 \ln \left(\frac{x-1}{x+4} \right) + c$$

$$b. \int_0^{\frac{\pi}{4}} x \sin 2x dx$$

$$u=x \quad u'=1$$

$$v = -\frac{1}{2} \cos 2x \quad v' = \sin 2x$$

$$= \left[-\frac{1}{2} x \cos 2x \right]_0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} \frac{1}{2} \cos 2x dx$$

$$= 0 + \left[\frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{4}$$

$$c. \int_0^{\ln 3} \frac{e^x \cdot e^x}{1+e^x}$$

$$\text{let } u = 1 + e^x$$

$$du = e^x dx$$

$$= \int_2^3 \frac{u-1}{u} du$$

$$= \int_2^3 1 - \frac{1}{u} du$$

$$= \left[u - \ln u \right]_2^3$$

$$= (3 - \ln 3) - (2 - \ln 2)$$

$$= 1 + \ln \frac{2}{3}$$

$$(\text{or } 1 - \ln \frac{3}{2})$$

d. let roots be α, α, β

$$\therefore 2\alpha + \beta = -a \quad (1)$$

$$\alpha^2 + 2\alpha\beta = 15 \quad (2)$$

$$\alpha^2\beta = 7 \quad (3)$$

sub (3) into (2)

$$\alpha^2 + 2\alpha\left(\frac{7}{\alpha^2}\right) = 15$$

$$\alpha^3 - 15\alpha + 14 = 0$$

$$\alpha = 1 \quad (\text{by inspection})$$

$$\therefore \beta = 7$$

$$\therefore a = -9$$

e. i) α, β, δ are solutions

$$\therefore \alpha^3 - 4\alpha + 5 = 0$$

$$\beta^3 - 4\beta + 5 = 0$$

$$\delta^3 - 4\delta + 5 = 0 \quad \text{adding}$$

$$\begin{aligned} \alpha^3 + \beta^3 + \delta^3 &= 4(\alpha + \beta + \delta) - 15 \\ &= 4 \times 0 - 15 \\ &= -15 \end{aligned}$$

ii) as $\alpha + \beta + \delta = 0$

$$(\alpha + \beta)^2 (\alpha + \delta)^2 (\beta + \delta)^2$$

$$= (-\delta)^2 (-\beta)^2 (-\alpha)^2$$

$$= (\alpha\beta\delta)^2$$

$$= (-5)^2$$

$$= 25$$

$$2. \quad a. \quad \int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta}$$

$$t = \tan \frac{\theta}{2}$$

$$\cos \theta = \frac{1-t^2}{1+t^2}$$

$$d\theta = \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{\frac{2dt}{1+t^2}}{2 + \frac{1-t^2}{1+t^2}}$$

$$= \int_0^1 \frac{2dt}{3+t^2}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right)$$

$$= \frac{\pi}{3\sqrt{3}}$$

$$b. \quad \int \frac{2x}{x^2 + 4x + 5} dx$$

$$= \int \frac{2x+4}{x^2+4x+5} - \int \frac{4}{(x+2)^2+1}$$

$$= \ln(x^2+4x+5) - 4 \tan^{-1}(x+2) + C$$

$$c. \quad I_n = \int_0^2 (x^3 - 8)^n dx$$

$$u = (x^3 - 8)^n \quad u' = 3n x^2 (x^3 - 8)^{n-1}$$

$$v = x$$

$$v' = 1$$

$$= \left[x(x^3 - 8)^n \right]_0^2 - 3n \int_0^2 x^3 (x^3 - 8)^{n-1} dx$$

$$= 0 - 3n \int_0^2 (x^3 - 8)(x^3 - 8)^{n-1} + 8(x^3 - 8)^{n-1} dx$$

$$= -3n I_n - 24n I_{n-1}$$

$$(3n+1)I_n = -24n I_{n-1}$$

$$I_n = \frac{-24n}{3n+1} I_{n-1}$$

$$d. \quad 1. \quad z^7 - 1 = 0$$

$$(z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$$

α is complex $\therefore \alpha - 1 \neq 0$

$$\therefore \alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$$

II. Sum of roots 1 at a time:

$$= \alpha + \alpha^6 + \alpha^2 + \alpha^5 + \alpha^3 + \alpha^4$$

$$= -1$$

Sum of roots 2 at a time

$$= (\alpha + \alpha^6)(\alpha^2 + \alpha^5) + (\alpha + \alpha^6)(\alpha^3 + \alpha^4) + (\alpha^2 + \alpha^5)(\alpha^3 + \alpha^4)$$

$$= \alpha^3 + \alpha^6 + \alpha^8 + \alpha^{11} + \alpha^4 + \alpha^5 + \alpha^9 + \alpha^{10} + \alpha^5 + \alpha^6 + \alpha^8 + \alpha^9$$

$$= \alpha^9 + \alpha^6 + \alpha + \alpha^4 + \alpha^4 + \alpha^5 + \alpha^2 + \alpha^3 + \alpha^5 + \alpha^6 + \alpha + \alpha^2$$

$$= 2(\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6)$$

$$= 2(-1)$$

$$= -2$$

sum of roots 3 at a time

$$= (\alpha + \alpha^6)(\alpha^2 + \alpha^5)(\alpha^3 + \alpha^4)$$

$$= \alpha^6 + \alpha^7 + \alpha^9 + \alpha^{10} + \alpha^{11} + \alpha^{12} + \alpha^{14} + \alpha^{15}$$

$$= \alpha^6 + 1 + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + 1 + \alpha$$

$$= 2 - 1$$

$$= 1$$

$\therefore$ equation is

$$x^3 + x^2 - 2x - 1 = 0$$

$$\therefore a=1, b=-2, c=-1$$

$$3. \quad a. \quad \int_2^4 \frac{dx}{x \sqrt{x-1}}$$

$$\text{let } u = \sqrt{x-1}$$

$$du = \frac{dx}{2\sqrt{x-1}}$$

$$u^2 = x-1$$

$$= \int_1^{\sqrt{3}} \frac{2 du}{u^2 + 1}$$

$$= 2 \tan^{-1} u \Big|_1^{\sqrt{3}}$$

$$= 2(\tan^{-1} \sqrt{3} - \tan^{-1} 1)$$

$$= 2\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$$

$$= \frac{\pi}{6}$$

$$b. \quad \int \sin^4 x \cos^3 x \, dx$$

$$= \int \sin^4 x (1 - \sin^2 x) \cos x \, dx$$

$$\text{let } u = \sin x$$

$$du = \cos x \, dx$$

$$= \int u^4 - u^6 \, du$$

$$= \frac{1}{5} u^5 - \frac{1}{7} u^7$$

$$= \frac{1}{5} \sin^5 x - \frac{1}{7} \sin^7 x + C$$

$$c. \quad 2 \text{ roots are } 1+2i, 1-2i \quad \therefore 1+2i + 1-2i = 2$$

$$(1+2i)(1-2i) = 5$$

$$\therefore P(x) = (x^2 - 2x + 5)(ax + b)$$

$$\text{constant term} \Rightarrow 5b = -15$$

$$b = -3$$

$$\therefore P(x) = (x^2 - 2x + 5)(ax - 3)$$

$$\text{but } P(2) = 5$$

$$\therefore 5(2a - 3) = 5$$

$$a = 2$$

$$\therefore P(x) = (x^2 - 2x + 5)(2x - 3)$$

d. $p(x) = x^4 + x^3 - 3x^2 - 5x - 2$

$$p'(x) = 4x^3 + 3x^2 - 6x - 5$$

$$p''(x) = 12x^2 + 6x - 6 \quad \text{triple root is root of } p''(x)$$

$$6(2x^2 + x - 1) = 0$$

$$6(2x-1)(x+1) = 0$$

$$x = \frac{1}{2}, -1$$

$$p'(-1) = 0$$

$$\therefore -1 \text{ is triple root}$$

$$\therefore -1 + -1 + -1 + \alpha = -1 \quad \text{sum of roots}$$

$$\alpha = 2$$

$$\therefore \text{Solutions are } -1, -1, -1, 2$$

e. let $y = \frac{1}{x+1} \Rightarrow x = \frac{1}{y} - 1$

$\therefore$ required polynomial is

$$p(y) = \left(\frac{1}{y} - 1\right)^3 + p\left(\frac{1}{y} - 1\right)^2 + q$$

$$= \frac{1}{y^3} - \frac{3}{y^2} + \frac{3}{y} - 1 + \frac{p}{y^2} - \frac{2p}{y} + p + q$$

$$= 1 - 3y + 3y^2 - y^3 + py - 2py^2 + py^3 + qy^3$$

$$= (p+q-1)y^3 + (3-2p)y^2 + (p-3)y + 1$$

$$= y^3 + \frac{3-2p}{p+q-1}y^2 + \frac{p-3}{p+q-1}y + \frac{1}{p+q-1}$$

$$\therefore a = \frac{3-2p}{p+q-1} \quad b = \frac{p-3}{p+q-1}$$

$$c = \frac{1}{p+q-1}$$